Diversity-Enhanced Evolutionary Optimisation of 3D Agents in Framsticks

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Introduction

Problem. Evolving a Framsticks creature whose **centre of gravity (COG)** must move in a set way (follow a line, bounce, jump, travel) means optimising its body and brain together. The search space is rugged, so evolutionary algorithms **converge early on one body plan**.

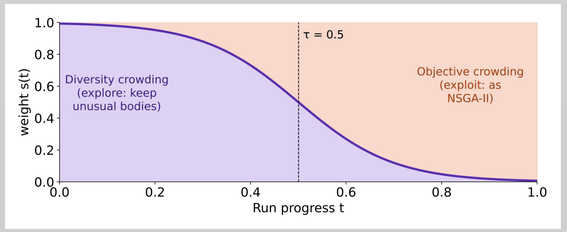
Gap. DE-NSGA-II [1] keeps solutions diverse by crowding on **genetic or structural dissimilarity**, but was only tested on simple tasks in one flat world, and it lost some fitness.

Aim. Test DE-NSGA-II and our new **Adaptive DE-NSGA-II** on tasks and worlds **modelled on the GECCO Automated Design Competition** [2] (its real settings are hidden), against NSGA-II [3] and the organisers' **Simple EA**, with the **f0** and **f1** encodings.

Approach & Methods

Five algorithms, one NSGA-II engine

- **NSGA-II** (baseline): objective-space crowding
- **DE gene**: crowd by genotype (Levenshtein) distance
- **DE graph**: crowd by structural body distance
- **DE hybrid**: weighted mix of gene and graph
- **Adaptive DE-NSGA-II (new)**: diversity crowding early, fitness crowding late



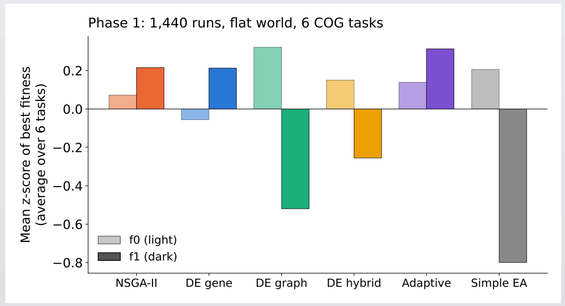
Three phases, 3,780 runs

- **Phase 1:** 6 algorithms × f0/f1 × 6 COG tasks × 20 seeds
- **Phase 2:** sensitivity of population, tournament, τ, w
- **Phase 3:** 14 reconstructed competition-style worlds + 4 held-out worlds

Equal budget: 30,000 offspring per run. Kruskal–Wallis + Dunn, Mann–Whitney, Wilcoxon (α = 0.05). CHPC Lengau, ≈1,860 CPU-hours.

Results I: Fitness

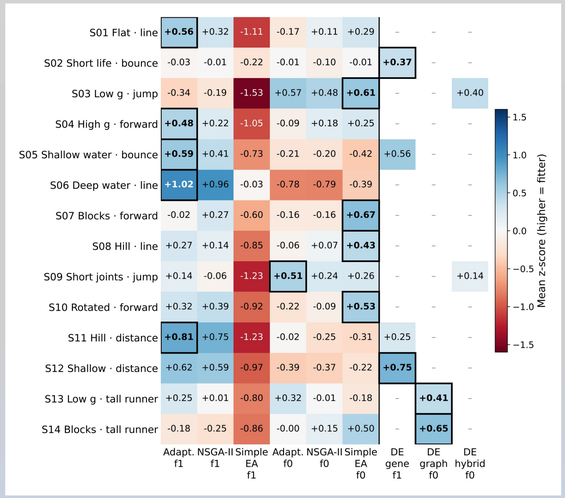
Phase 1: no single winner



- No DE variant was **significantly fitter than NSGA-II** on any task.
- **Adaptive** beat DE gene across tasks (p = 0.034), but **not NSGA-II** (p = 0.42).
- **Encoding mattered more:** f1 for movement, f0 for jumping.

Task	Best algorithm (mean)
Straight line	Adaptive f1
Sinusoid bounce	DE gene f1
Forward velocity	NSGA-II f1
Jump height	DE hybrid f0
Distance (fn 3)	DE gene f1
Tall runner (fn 4)	DE graph f0

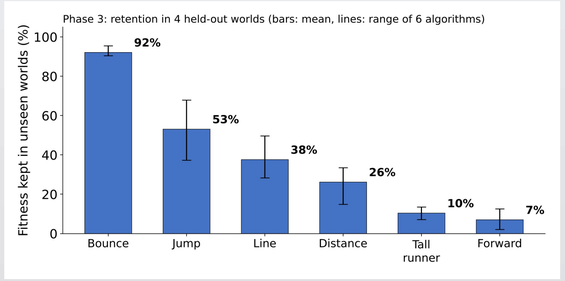
Phase 3: 14 reconstructed competition worlds



- Adaptive f1 best in **5/14** settings; Simple EA f0 best in **4/14**.

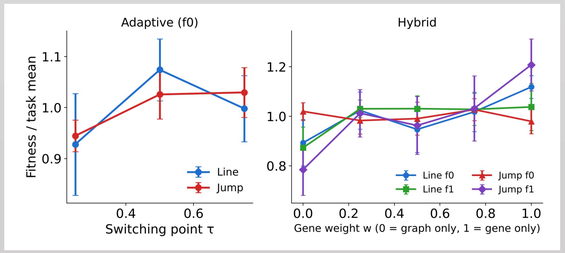
Results II: Robustness

Agents generalise poorly



- Only **32–40%** of fitness kept in unseen worlds; walking gaits broke.
- Crowding rule: **no significant effect**. f0 kept slightly more (p = 0.004).

Phase 2: settings barely matter



- No sweep significant (p ≥ 0.145); early switching (τ = 0.25) and graph-only (w = 0) weakest.

Diversity of the 20 best solutions

Variant vs NSGA-II	More diverse	Less diverse
DE gene & graph (Phase 1, own measure)	7	2
Adaptive (Phase 3, both measures)	15	29

Counts of significant differences (Kruskal–Wallis + Dunn, as in [1]). Fixed DE crowding keeps the best solutions more diverse; Adaptive ends more converged because it exploits late.

Key Findings

1. Diversity ≠ fitness

Fixed DE-NSGA-II kept the best solutions more diverse, as in [1], but was **never fitter than NSGA-II**.

2. Adaptive is the most consistent

Explore early, exploit late **removed the fitness cost** of diversity. Best general algorithm with f1, but **not significantly** better than NSGA-II.

3. Encoding > crowding

f1 suits movement, f0 suits jumping; graph crowding is too slow for f1.

4. Simple EA is a strong baseline

Beaten on f1 in 13/14 settings, **never beaten significantly on f0**.

5. Generalisation is the open problem

Next: full competition budget, state-based schedule, multi-world training, a 2027 entry.

Acknowledgements

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Framsticks 5.5 via framspy; Python 3.12 framework with 58 unit tests.

Scan me

Code, results, the poster and an **animation of the evolved creatures**.

github.com/KENJI-MASAMUNE/Optimizing-3D-Agents-Showcase

References

[1] M. Komosinski and A. Mensfelt, "Enhancing quality-diversity optimization through domain-specific dissimilarity as crowding distance," GECCO 2025, pp. 898–906.

[2] Automated Design Competition at GECCO 2024–2026, framsticks.com/gecco-competition.

[3] K. Deb et al., "A fast and elitist multiobjective genetic algorithm: NSGA-II," IEEE TEC, 6(2), 2002.